

Design and Implementation of a Single Pulse Active Radar System Optimized for Target Classification



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Declaration

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Acknowledgements

At the start of my engineering degree, a friend of my parents, who happens to be an engineer himself, heard that I had been accepted into the electrical engineering faculty at UCT. I asked him for any advice that he might have for me, anything that could potentially help me through the four years that lay ahead of me. He said one thing to me, ‘Make a group of close engineering friends, stick together, and you’ll get each other through’. This was some of the best advice that I have ever received. To everyone in the engineering friend group, getting to know everyone over the last few years has been one of the coolest experiences in my life. Thank you for all the memories made. I cannot wait to see what interesting things everyone gets up to in the coming years.

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Abstract

Every day without fail, all around the world, there are thousands of weather balloons that are launched into the atmosphere for meteorological observation and research applications. These weather balloons are all comprised of a latex balloon that is filled with helium, and a radiosonde unit that hangs below the balloon itself. This radiosonde module plays a very important role in the measurement of atmospheric conditions within which the balloon resides, as it contains the numerous electrical components that give it the ability to measure characteristics of the atmosphere and the position in which the radiosonde finds itself in, as well as transmit those readings back to a ground station.

Currently, there are only a handful of companies around the world that design, manufacture, and sell radiosonde modules. These companies essentially have a monopoly on the entire market. One of the hindrances to companies/individuals designing and manufacturing their own radiosonde modules is the complicated process of transmitting the recorded information back to a ground station. Since the companies that manufacture the radiosonde modules have closed designs on all of their components, information is not widely available as to the exact inner workings of their designs for their transmission systems, other than the broad modulation schemes. It is therefore of paramount importance that alternative transmission schemes to the schemes that are employed by the few companies that manufacture radiosondes are explored if alternative designs to the mass-manufactured models are desired.

This report outlines research into an alternative transmission system based on the [Long Range \(LoRa\)](#) communication system. While LoRa is still a proprietary communication system, it is widely available due to its popularity among hobbyist communities, as well as its expanding footprint within the [Internet of Things \(IOT\)](#) community of devices. Reasons for popularity within these communities include the long-range transmission capabilities of the communication protocol, as well as its low cost, which makes it an excellent choice for the disposable radiosonde modules that are used for atmospheric observations.

A transmission and receiver system was designed, simulated, and built, based around the LoRa transmission protocol. Tests were then performed on the system to determine the optimal LoRa parameters for the usage of the protocol in radiosonde modules. From testing and simulation data, the best bandwidth was determined to be 62.5kHz, and the best spreading factor was determined to be 11.

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Orthogonal Frequency Division Multiplexing (OFDM)

Acronyms

ADC Analog to Digital Converter

AEL American Electric Laboratories

ASR Airport Surveillance RADAR

AT&T American Telephone and Telegraph Company

ATPs Acceptance Test Protocols

BER Bit Error Rate

CDS Cobham Defense Systems

DAB Digital Audio Broadcasting

FDM Frequency Division Multiplexing

FMCW Frequency Modulated Continuous Wave

GPS Global Positioning System

GPU Graphics Processing Unit

IOO Illuminators of Opportunity

IOT Internet of Things

LoRa Long Range

NI National Instruments

OFDM Orthogonal Frequency Division Multiplexing

PAPR Peak to Average Power Ratio

QAM Quadrature Amplitude Modulation

RADAR Radio Detection and Ranging

Radcom RADAR Communications

RRSG Radar and Remote Sensing Group

SDR Software Defined Radio

SNR Signal to Noise Ratio

TDM Time Division Multiplexing

TDMA Time Division Multiple Access

Chapter 1

Introduction

In our ever evolving technological world, new ways of solving problems are being developed at a rapidly increasing rate. There is much to be said about developing brand-new methods in solving existing problems, however, in this report I propose the combination of two, fairly well-thought-out existing technologies, to try and solve a useful problem in the world of [Radio Detection and Ranging \(RADAR\)](#).

This report outlines the thought process, design, and implementation of a single pulse RADAR system, combined with an [OFDM](#) based signal, to achieve the goal of target classification from a scene. RADAR systems in today's world can perform certain tasks extremely well (in the right conditions), however, there still seems to be a lack of RADAR systems that can identify/classify what objects might be sound in a scene of interest, without the utilization of complex image processing techniques.

In this paper, I will outline the process that was gone through to build a working prototype system, that uses the combination of a single pulse RADAR, and an OFDM based signal, to try and classify objects in a scene.

This report will consist of the following loose structure. Firstly, a literature review of the related works to do with the fundamentals of the project, namely, single pulse RADARs, OFDM, and the mixture of the two. Next, a theory section that concerns the workings behind the project will be presented. The reason a separate theory section is included in this report is due to necessity of having a section dedicated towards fundamentals behind the system which I am aiming to design. Next, the design of the surrounding system to transmit and receive data, process the data, and analyze the results is presented. Following this, an experimental design is laid out and explained in detail. This is followed by the performing of real world testing which is then analyzed in the results section. Lastly, there will be some discussion of results and further steps that could be implemented.

1.1 Background

Throughout the years, humans have sought out better ways to sense and detect objects that exist in the physical world, using methods that differ from our natural senses. Naturally, RADAR technology has unlocked the ability for humans to be able to see the world around us with a new perspective. This ability to be able to see objects and more specifically, to see information such as position and velocity about these objects has helped with the development of devices and services that are of paramount importance to society, be it from a defense perspective, a safety perspective, or even in the operation of consumer products. All the current use cases of RADAR do however lack the ability to determine

exactly what object is being seen by the overall RADAR system. That is where the use of the OFDM waveform in a RADAR system becomes useful.

1.2 Problem Statement

With the current state of pulsed (and single pulse) RADAR systems worldwide, the ability to infer the following about point targets (and multiple point targets for that matter) becomes possible, firstly, the distance that an object is from the receiver and transmitter setup can be inferred, then, the direction (elevation and azimuth) of the object can be inferred via multiple different methods, although in a simple pulse or pulsed RADAR, direction calculations usually come from the direction that an antenna is pointing, lastly, one can determine the relative velocity of an object. While these inferred properties of objects is useful, modern system break down when it comes to target classification. Target classification would greatly benefit society to help determine more precisely what targets in a RADAR scene are. There exists a gap in the research in regard to target classification abilities of simple RADAR systems.

1.3 Objectives

The main objective of this project is to demonstrate the design process that has gone into the realization of a simple single pulse RADAR system that can simultaneously act as a normal RADAR system which can determine the distance and rough direction of an object, as well as be able to perform target classification on objects contained within a scene. The goal of target classification should be done by implementing a custom [OFDM](#) based transmit and receive tool chain. This OFDM based system should have the capability to transmit and receive data, perform simple RADAR tasks, and be able to perform the desired classification objectives. This system will not be real-time, as real-time implementation of RADAR processing techniques in the software domain take momentous amounts of compute.

1.4 Scope and Limitations

The RADAR space is an extraordinarily broad and complex space that has much research emanating from multiple institutions every year. Following on from this, the same can be said for communications systems, especially when dealing with modulation techniques such as [Quadrature Amplitude Modulation \(QAM\)](#) and [OFDM](#). This allows for a research space that can easily become overly complicated in nature.

The scope of this project has been set to not include design of RADAR hardware specifically. This is because design of RADAR hardware is a complex task that would probably encompass an entire masters project in and of itself. Therefore, the RADAR hardware portion of the project will be handled by a [Software Defined Radio \(SDR\)](#). This extrapolates the complex RADAR hardware design aspect away from the project and allows me to focus on the data collection and processing aspect of the project.

Chapter 2

Literature Review

In the following chapter, I aim to investigate the relevant literature regarding the implementation of a single pulse radar optimized for target classification using off the shelf SDR components. I aim to understand what has been done in the past, with the equipment I aim to use, and how this might tie into my current attempt at implementing such a system with similar components. The first literature to explore is the work that pertains to using SDRs in a radar system, specifically the SDRs that I have access to, the USRP N210s. Next, one must explore the use of OFDM based signals in radar systems. And lastly, I will explore radar target classification and the various avenues researchers use to try to achieve target classification.

2.1 SDR Based Radar Systems

Since the goal of SDRs is to make researching and building RF based systems more achievable for more people, much has been written and published on the topic of SDRs. Since radar is a popular area of interest for many researchers, there are various different papers pertaining to SDRs being used in real world radar systems. Examples of such usage include, but are not limited to, the following. In [2], the authors propose a framework wherein a combination of SDR technology and [Long Range \(LoRa\)](#) are used to try to help the efficiency of search and rescue operations. The SDR is used to broadcast an [Frequency Modulated Continuous Wave \(FMCW\)](#) waveform. They go on to evaluate the efficacy of the approach through experimental implementation. In [3], the authors propose and test a passive ionosphere radar built around the USRP N210 SDRs. This project utilizes a dual USRP setup, which is interesting as it is similar to the proposed setup for my project. In [4], we are told about the implementation of an FMCW based radar designed for use as a weather surveillance instrument. Once again, this radar is built around the Ettus N210 SDRs.

As can be seen, the use of SDRs in research for creating and implementing radars of various sorts is fairly common.

2.2 OFDM Based Radar Systems

In [5], the author explains in detail a theoretical implementation of an OFDM based radar system. He argues that the combination of the communication signal and a radar system could unlock capabilities including but not limited to, having multiple radars examining a scene and all communicating amongst each other.

2.3 Current Target Classification Abilities of Radars

Chapter 3

Theory

Broadly speaking, the design and implementation of my system fell into two distinct categories, the theoretical side, and the practical side. To be able to fully understand the practical implementation of my single pulse RADAR system, I feel that it is important that one must first have a firm understanding of the theoretical principles that are being utilized by the RADAR system to achieve the desired goals of the project, before moving onto the practical system design outlined in chapter 4. For this reason, I have included this theory section, which will dive into the important components that have a need to be understood in my project. This will hopefully give you the reader the ability to better understand the sections that follow, as a solid grounding will have been set up.

3.1 Software Defined Radios

Not too many years ago, when a team was working on a RADAR or radio system, they would be split up into various different boxes. Someone would work on the RF frontend, perhaps another on the amplification, and another on the [Analog to Digital Converter \(ADC\)](#). This was the old way of doing things [6]. Today however, with the speed at which new systems, specifically in the realm of communications systems, have been progressing, it has become necessary to come up with better ways to prototype with radio frequency based systems. There has been an explosion in SDRs that give engineers the ability to implement baseband functionality in software, while also having the ability to control the RF frontend via software. ‘An SDR system is a complex device that performs several complicated tasks simultaneously in order to enable the seamless transmission and reception of data’ [6].

3.1.1 Theory of Operation

To understand how an SDR works, let’s imagine a simple SDR system, with the humble goal of capturing some raw analog data from the RF domain, and converting that into the analog baseband domain, and then from that into IQ samples, which one can work with in a software package. There are three main components that go into this in order for this to work, which I have listed below:

1. **RF Frontend:** This is the system’s ability to sample

3.1.2 Bandwidth**3.1.3 Sampling Rate****3.1.4 Tuning Frequency****3.2 Orthogonal Frequency Division Multiplexing****3.2.1 Basic Operating Principles****3.2.2 Data Modulation****3.2.3 Channel Estimation****3.2.4 Equalization****3.3 Principal Component Analysis**

In this chapter, I aim to explore the existing literature concerned with building classification radar systems using OFDM, and SDRs. Then I delve into [OFDM](#), followed by [SDR](#) and its working and uses within the world of RADAR. Then, I look into OFDM based AR, and finally, I look into OFDM based classification RADAR. My aim for this review of the literature is to add context to the broader paper, and hopefully give you, the reader, a more in depth understanding of the core concepts one has to understand to fully grasp the overall workings of the project. Especially as we move towards the latter parts of the project, where classification is involved, it is important to grasp the fundamentals.

I will separate the literature review into distinct sections, where I will explore the history behind each concept and the theory involved within. I will also explore and analyze the advantages and disadvantages of each topic and then finally, provide a critique of the literature towards the end.

3.4 Classical RADAR**3.4.1 Brief History**

The fundamental idea behind RADAR systems and the technology itself were first theorized during the classical electromagnetic radiation experiments of the 1880s performed by Heinrich Hertz [7]. These experiments set out to prove the previous theoretical work achieved by James Clerk Maxwell. Maxwell came up with the formulae that described electromagnetic fields, and he hypothesized that this formula would apply to both light and radio waves since they were both examples of electromagnetic radiation just at vastly different frequencies. This has obviously since been proved, with visible light roughly falling into the 400THz to 700THz frequency range. The work done by Maxwell led to the discovery that electromagnetic waves can be reflected by metallic objects and refracted by a dielectric medium, just like light can [7]. This work did not go unnoticed and a patent for an ‘an obstacle detector and ship navigation device’ was issued in 1904 to Christian Hülsmeyer, a German engineer [7]. Work on RADAR then slowed down due to a lack of interest from governmental authorities until the early 1930s, when the development of long-range bombers prompted the major powers of the world to develop technologies which could detect these hostile aircraft, which would be useful for the coming second

world war.

3.4.2 RADAR Workings

At its core, a RADAR is defined as an instrument that can detect surrounding objects using radio waves. To achieve this, there are multiple different setups of RADAR systems, which will be discussed later. These RADAR setups however all rely on the following fundamental laws of physics which are listed below [8]:

1. **Reflection of electromagnetic waves:** This rule describes how a radio wave will be reflected back towards a transmitter of said radio wave if the object is placed directly in front of the transmitter. This reflection indicates that the object is in front of the transmitter.
2. **Electromagnetic energy travels through air at a constant speed:** This rule describes how the electromagnetic waves transmitted by an antenna propagate through the transmission medium, usually air for our earth's use cases, at roughly the speed of light, c . This velocity c equates to $299,792,458 \text{ m.s}^{-1}$
3. **Electromagnetic energy normally travels through space in a straight line:** This describes how the energy transmitted by an antenna in a radar system propagates through space in a straight line. This is important because it means that the energy of the RADAR system can be directed in a specific direction by the antenna system, which means that the reflecting object's direction (azimuth and elevation) can be determined.

In a classic pulsed RADAR, essentially the most simple kind of RADAR that is used today, by using the above-mentioned physical properties, the distance, azimuth, and elevation of the object that is being observed by the radar system can be determined. The best example of this sort of RADAR system is an airport RADAR. An [Airport Surveillance RADAR \(ASR\)](#) is used at airports to determine the presence and position of aircraft in the vicinity of an airport. Typically, ASR systems are capable of reliably detecting and tracking aircraft at altitudes below 25,000 feet (7,620 meters) and within 40 to 60 nautical miles (75 to 110 km) of the airport [9].

Classical RADAR systems can be separated into a few different categories [1]. Broadly, you get primary and secondary RADARs that are separated into imaging and non-imaging RADARs. An imaging RADAR is one which desires to draw an image, such as a weather RADAR, and a non-imaging RADAR is one which does not draw an image but rather infers information about a target based on processing techniques. Primary RADARs rely on echos from a target that come from it's own transmitted pulse, whereas a secondary radar relies on a RADAR response signal from a target. Sometimes, the ASR systems mentioned previously have secondary radar systems built-in which allow the aircraft being detected to broadcast vital information to aid with air traffic control. This is usually only possible for 'friendly' targets. The various categorizations of RADAR types can be seen in figure 3.1 below:

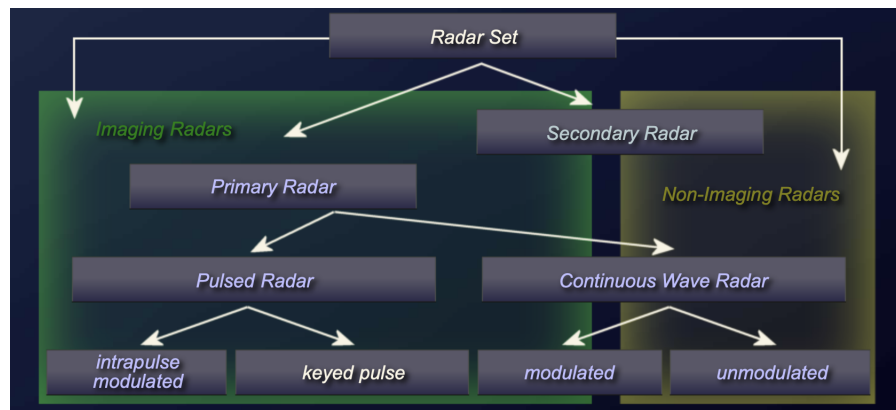


Figure 3.1: Classification of Radar Systems [1]

Note to self, I don't think I like this image

3.4.3 Advantages and Disadvantages of RADAR

Since RADAR utilizes the electromagnetic spectrum, there are distinct advantages concerning the use of RADAR systems [8]. The main advantage is that, unlike humans, RADAR systems can operate day and night, 24/7, over far distances, in good or bad weather. Compared to human eyes, this is obviously a lot better. Some other advantages of classic RADAR is that it can generally 'see' through insulators, and it can also distinguish the difference between fixed and moving targets [10]. Some disadvantages however, are that classical RADAR systems generally struggle to resolve targets that are very close to the transmitter, and they cannot penetrate water very well [10]. Classical RADAR systems also fail to distinguish between shapes of object that they are looking at.

3.5 OFDM

3.5.1 Brief History

The general working principles, and the mathematics required for OFDM were originally described by Robert Chang in 1966 while he was working at Bell Labs [11]. While the idea of OFDM was smart, it didn't become popular until sufficiently cheap microprocessors entered the market [11]. OFDM was never implemented in commercial communication channels until the appearance of mass market applications coincided with the availability of efficient software and electronic technologies [12].

3.5.1.1 Previous Transmission Techniques

Before OFDM there was [Frequency Division Multiplexing \(FDM\)](#). FDM system innovation was driven out of necessity due to telegraph companies hoping to increase the amount of users that could utilize their transmission lines to increase profits from their systems [12]. Transmission companies were in search of a method to have multiple messages being sent on the same transmission line at simultaneously. Other methods had been used such as [Time Division Multiplexing \(TDM\)](#), and [Time Division Multiple Access \(TDMA\)](#). These methods were however limited by burst speed which was itself limited by inter-symbol interference. Around the 1920s, FDM became the main carrier system for telephone

companies. The first major company to deploy the system was [American Telephone and Telegraph Company \(AT&T\)](#). This initial deployment of the system used 8kHz subchannels with 4kHz allocated to each channel. This form of transmission system later developed into systems that had more subcarrier channels with less subcarrier spacing. There were however some drawbacks to FDM, which are highlighted below:

1. **Waste of frequency:** Due to the large subcarrier spacing, FDM wasted a lot of precious frequency spectrum to these so-called ‘guard’ bands on either side of the carrier. As the frequency spectrum became more crowded throughout the years, these guard bands became a liability due to demand for spectrum being to high, and the guard bands were not efficiently using the spectrum.
2. **Using different modulators for the various sub-channels:** In the old systems, each sub-channel had a separate modulator in the up-mixing process. This multiplication process became tedious once there were multiple sub-carriers.

To compensate for this inefficient use of the frequency spectrum, OFDM was introduced. What made OFDM different to FDM was that each subcarrier overlapped some of the other carriers in signal. This was able to be done because each carrier was spaced in such a way that it was orthogonal to the other carriers in the signal [12].

3.5.1.2 OFDM

As mentioned before, [OFDM](#) is a transmission method that uses the spectrum in a more efficient manor to transmit as much information as possible over a single transmission line. The transmission of information is commonly done using various modulation schemes such as QAM encoding that breaks information into bit chunks that are sent in parallel over multiple channels [13]. The OFDM transmission scheme builds upon FDM where multiple channels are created over a specific bandwidth, and bits of information are sent on each channel in parallel. The main difference between FDM and OFDM is therefore the orthogonality of the OFDM signal. I have created a crude example below in figure 3.2. In fig 1.2a, a crude mockup of an OFDM signal can be seen in the frequency domain. As can be seen, all the peaks of the OFDM waveform have zero interference with any other part of the signal. This is due to the various carriers being orthogonal to each other. Conversely, with the FDM waveform, the peaks have significant interference from the other channels since the channels are not spaced orthogonally in regard to each other.

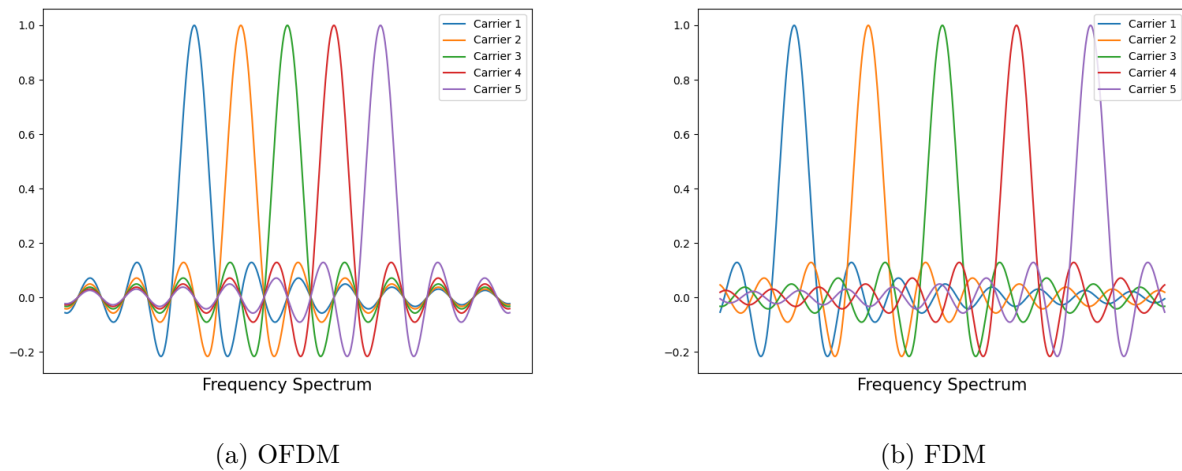


Figure 3.2: OFDM and FDM Waveforms

I want to touch on the idea of orthogonality. In practice, all it means is that in a signal with multiple sinusoidal carriers, each carrier has a frequency that is a whole number multiple of some fundamental frequency [11]. The time domain voltage signal of an OFDM signal can be described by equation 3.1 below:

$$v(t) = \sum_{k=0}^{N-1} X_k e^{\frac{j2\pi kt}{T}}, \quad 0 \leq t < T \quad (3.1)$$

N = Number of subcarriers

where X_k = Data symbols in the frequency domain

T = OFDM symbol time (s)

The subcarrier spacing of $\frac{1}{T}$ ensures carrier orthogonality over the OFDM frame, or symbol. In practice, real world implementations of OFDM signals will have vastly more complex components added on concerning error correction and data modulation techniques in general. The base signal however will still revolve around the idea of orthogonal carriers.

3.5.2 Advantages and Disadvantages of OFDM

As can be seen in the above sections, OFDM is a very useful and versatile transmission technique that is used in a variety of real world applications. While OFDM is a widely used and very good transmission technique, it does come with some pros and cons [14]. The biggest pro for OFDM, particularly with its use within the telecommunications industry, is the spectral efficiency that can be achieved. Real world OFDM systems are also resistant to multipath fading due to the cyclic prefix that these systems implement. Modern OFDM systems can allow for easy equalization with embedded pilot symbols. Equalization is much harder to achieve in a single carrier system. Some cons of OFDM systems however include the following. Firstly, an OFDM transmission can have a high [Peak to Average Power](#)

Ratio (PAPR). This is due to an OFDM signal's noise-like amplitude with a large dynamic range [14]. Another problem is that OFDM systems are susceptible to carrier frequency offset due to local oscillators not being able to perfectly operate at the same frequency on the transmitter and receiver.

3.6 OFDM based RADAR

3.6.1 Brief History of OFDM based RADAR

3.6.2 Implementation of OFDM based RADAR

There is an ever evolving area of RADAR research that involves the combination of RADAR systems and telecommunication systems. This is colloquially known as **RADAR Communications (Radcom)**. Radcom aims to combine radar and telecommunications because this could potentially lead to system that are more spectrally efficient, and lightweight. In essence, one could imagine a RADAR system that can communicate with other base stations, as well as performing usual RADAR tasks, all from a single system. An attempt at creating a system like this was attempted by Dr Stephen Paine and Grant Norrie in [15]. They set out to create a RADAR system that utilizes the previously mentioned OFDM signal structure in a radcom setup. They wanted to base the system around an OFDM signal because of the ever-increasing popularity of OFDM signals in communication networks. They mention that communication signals are often used in passive RADARs, by exploiting **Illuminators of Opportunity (IOO)**. An IOO is a transmitter that is not intended for RADAR purposes, but rather other applications, with the most popular being telecommunications purposes. They pose the question, if a passive RADAR system can use existing communication signals to detect objects, why can an active system not do the same?

They explored the use of the OFDM based **Digital Audio Broadcasting (DAB)** modulation scheme as the backbone of the system. Part of the reason for this, is the fact that DAB is a very popular and well-defined. They designed and documented an entire toolchain that consisted of the following parts.

1. **Signal Generation Chain:** This entailed the building of the signal that was to be transmitted by their receiver.
2. **Hardware Testbed:** This was the physical hardware that was used to implement the theoretical in the real world. This consisted of dual Ettus USRP N210 SDRs.
3. **Radar Processing Chain:** Everything to do with processing of the signal on the reception side of things to verify the operation of the system as a RADAR. The main objective of this toolchain was to generate a range-doppler graph to visualize the output of the overall RADAR system.
4. **Communications Processing Chain:** This toolchain was in place to verify and perform the necessary operations to facilitate a working communication link between the transmitter and the receiver.

The SDR platform had specific configuration characteristics that were used for the tests performed by [15]. These settings are summarized in table 3.1 below:

Parameter	Value
Tx Gain	0 dB
Rx Gain	0 dB
Tx Bandwidth	1.1 MHz
Rx Bandwidth	1.1 MHz
Sampling Rate	5 MHz
Carrier Frequency	2.45 GHz

Table 3.1: SDR Toolchain Parameters

Multiple tests were performed to test the capability of the system. These tests were a communication test, a static range test, and a range-doppler test, which involved a moving target. The communication test was fairly basic, and it aimed to eliminate antenna/system feed through, and then measured the [Bit Error Rate \(BER\)](#) upon transmission from the transmit antenna to the reception antenna from a distance of 5 meters away. The BER was confirmed to be 0% and then the test was deemed to be a success.

Then a static range test was performed with an object 100m away from the RADAR. After pulse cancellation with adjusted dynamic range, and clutter map removal, a range-doppler map was generated as shown below in figure 3.3:

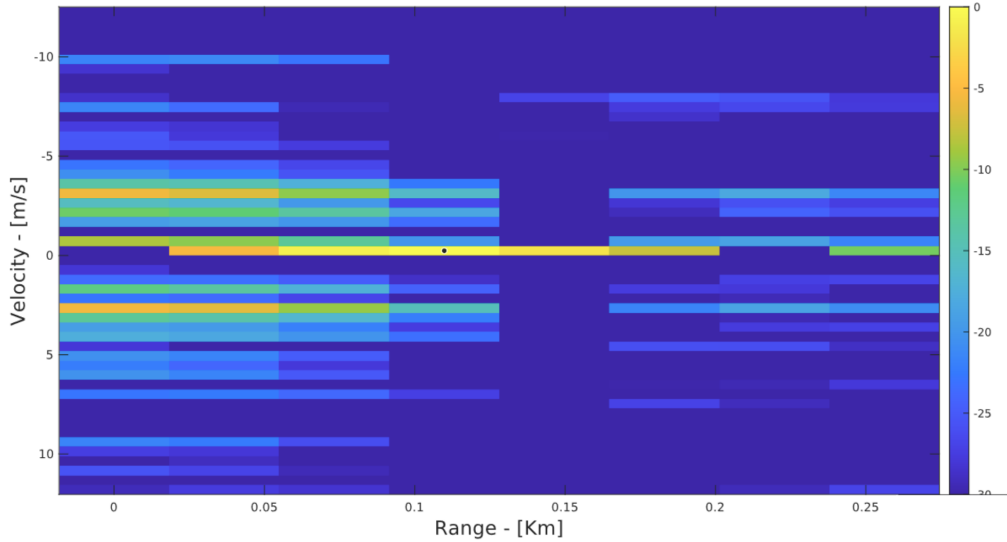


Figure 3.3: Range-doppler map of object at 100m

This figure showed that the object was detectable at 100m, specifically, the object could be seen in the 109m range bin. Lastly, a moving target test was performed, which was able to successfully detect a moving car. The ground truth of the moving object was as follows, it was 30m away from the RADAR and was moving at a velocity of 7.2 m.s^{-1} and 7.0 m.s^{-1} as measured by a [Global Positioning System \(GPS\)](#) and the odometer respectively. The measured range and velocity was determined to be 36m and -7.4 m.s^{-1} . The negative velocity is due to the object moving away from the RADAR system.

In [16], Gabriel Lellouch

3.6.3 Advantages and Disadvantages of OFDM based RADAR

3.7 Critique of Literature

Notes to self

...In Stephen Paine and Norrie paper, the communication test is quite trivial...

Chapter 4

System Design

In the following chapter, I will outline the exact thought process that went into designing my radar system. In order to make my explanation of the design process easier, and for continuity's sake, I have separated the overall system into three distinct modules. These modules all overlap in certain ways, however, it will be easier to explain the design process behind each module while assuming no overlap. The three modules are the following, firstly, a range radar module. This is the bare-bones part of the overall system that can determine the distance that objects are away from the transmitter and receiver. Secondly, there will be a part of the system that involves communications. This is the communication module. This is the part of the overall system that will demonstrate the ability of the system to perform basic operations as a communication system that sends data between and transmit and receive. Thirdly, there will be a classification module. This classification module will involve the prediction pipeline that is used to classify different shapes of objects in the frame of the radar.

This chapter aims to outline the thought process that went into the design of each of the modules highlighted above. Before any explanation of design thinking however, a requirement and specification analysis is outlined below.

4.1 Requirement and Specification Analysis

It is important to list out the specific system requirements and specifications for the overall system and the various submodules. This makes it easier to decide on the design decisions that must be made, and also makes designing of the validation, and the overall testing procedure easier.

4.1.1 System Requirements

1. **Overall System:** The main requirement of the entire system is that it be designed with widely available components, either consumer or scientific. Naturally, some scientific equipment will be fairly expensive, however, as long as this equipment is readily available, then that is adequate. The system should be able to seamlessly switch modes of operation so as to highlight the various operational capabilities of the three separate modules.
2. **Range Radar Module:** The range module should be able to detect the distance of basic objects positioned in a scene. It should be able to determine the distance of these object as accurately as possible within the bounds of the hardware and software implemented.
3. **Communication Module:** The communication module should be able to perform basic data

transfer via RF signals sent between the transmission and receive section of the overall system. This communication system does not have to include any advanced error correction, however, basic channel equalization must be implemented to make recovery of basic signals possible.

4. **Classification Module:** The classification module should be able to observe a scene, and be able to determine the difference between basic shapes placed in a scene. The basic shapes in question will be limited to objects present within the UCT radar group's possession. The classification pipeline should be able to classify objects that are stationary and moving, and as well as this it should be able to classify multiple objects within a single scene.

4.1.2 System Specifications

1. Overall System:

- (a) In sticking to the aforementioned requirements, the first specification for the overall system is that all components, consumer and/or scientific should be available through two independent suppliers.
- (b) Secondly, the overall system should have three distinct modes of operation. Each mode of operation should take no less than 20 seconds to switch to, and finally, each mode should seamlessly integrate with the other modes, so as not to degrade the system's stability, meaning no crashes and data corruption etc.

2. Range Radar Module:

- (a) The range radar should have a minimum range resolution of below 20m. This has been chosen as a good starting point for this radar because this is a single pulsed radar, not a **FMCW** radar. FMCW radars typically has much better range resolution capabilities, and does not rely on simple pulse compression in determining the distance of objects away from the transmitter and receiver, which this range radar does.
- (b) The radar should be able to reliably detect objects at distances greater than 10m.

3. Communication Module:

- (a) The communication pipeline should be characterized by a BER versus SNR, and should be able to achieve a near zero BER for high SNR values.
- (b) The communication system should be able to transmit data at a meaningful data rate that could be useful in a real world operation. To be able to operate in the real world, the module should be able to transmit at a rate of or above 1 000 000 bits/s, or 1 Mbps.

4. Classification Module:

- (a) The classification module should have shape classification abilities, or in other words, be able to classify or separate between distinctly differently shaped objects. The module should be able to correctly classify a minimum of three objects in any given scene.

- (b) The classification module should be able to handle basic motion, such as an object rotating along a single axis, and advanced motion, such as a target moving along multiples axes at various velocities.
- (c) The module should be able to handle a minimum of two targets with both at different distances away from the radar, while still maintaining the ability to classify said targets accurately.

4.2 Modules and Submodules

As mentioned previously, the overall project is broken down into three distinct modules. Breaking down a big project such as this one into different ‘black boxes’ helps dramatically with the design process by making daunting large parts of the overall project more manageable. The choice of broad submodules is shown below in fig 4.1.

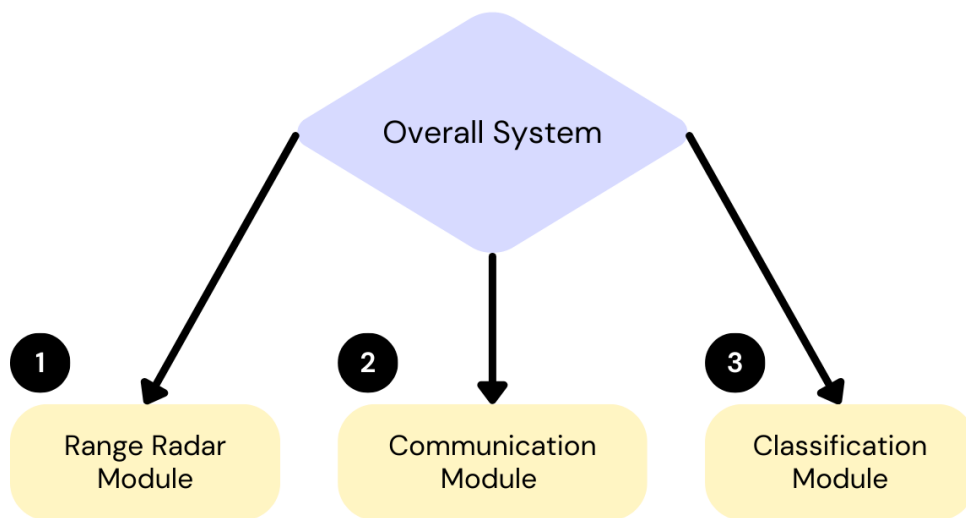


Figure 4.1: Modules and Submodules

Once an overall picture of the modules and sub modules can be outlined, decisions about the components contained within can be made.

4.3 Component Selection

The components of the system can be broken into two broad categories, namely, the hardware, and the software.

4.3.1 Hardware

Apart from the custom carrying rig, the hardware for this project is entirely made up of off the shelf components. The main components that fall part of the hardware category are the computer, which forms the basis of the processing unit of the radar, the SDR system, which contains all the required equipment to transmit and receive RF signals, the antennas, the carrying case, and the other periphery

equipment such as power management and external monitors, keyboard, and mouse. Following on from this, below is an explanation of each hardware component choice, and the reasoning behind each choice.

4.3.1.1 Computing Unit

The computing unit for this project needs to be powerful enough to handle development of the entire system. To keep things neat and tidy, I am doing all the development of the software, which will be addressed later, on the computing unit chosen for this project. Therefore, the unit must be able to compute vast amounts of data used within the project. This will be a CPU intensive task. As I plan on utilizing a multi-threaded approach to the processing pipelines of this system, a CPU with as many cores as achievable would be ideal. There is no plan to implement [Graphics Processing Unit \(GPU\)](#) integration into the project. Based on this information, I chose the GMKtec NUCBOX as the computer for this project. This choice of computer can be seen below in [fig 4.2](#).



Figure 4.2: GMKtec NUCBOX

This computer is a very well-rounded computing device, and has the following specifications outlined in [table 4.1](#) below.

CPU	AMD Ryzen 7 8845HS
CPU Clock Rate	5.10 GHz
Cores	8
Threads	16 (via Simultaneous Multithreading)
GPU	Radeon 780M RDNA3
RAM	32 GB DDR5-5600MHz
Storage	1TB M.2
Networking	2x Intel I226V 2.5 Gigabit Ethernet RJ-45 LAN Intel AX200 WiFi 6 Adapter Bluetooth 5.2
Peripherals	2x USB 4.0 Type-C 2x USB 3.2 Gen 2 Type-A 2x USB 2.0 Type-A 1x HDMI 2.1(8K@60Hz) 1x DisplayPort 2.1 (8K@60Hz) 1x 3.5mm Headphone Jack 1x Oculink (PCIe 4.0 x4)
Dimensions	132mm x 125mm x 58mm
Cost	R12 000.00 ¹

Table 4.1: GMKtec NUCBOX Specifications

As can be seen, this is a very well-rounded computing unit that offers many features one may find useful for a project such as this one. Particularly useful to this project is the fact that the unit offers 8 cores that can effectively handle 16 simultaneous threads when processing data. The unit also offers a large amount of RAM which can prove useful for processing large amounts of data. As will be explained in later sections of this chapter, a large amount of RAM is necessary for this project, because a large part of the processing for the system utilizes correlation calculations, which is a numerically intensive operation. Another selling point of this specific unit is the dual ethernet RJ45 ports. This proves useful for interfacing with the SDR devices, outlined later, while simultaneously accessing another network, possibly for data transfer purposes, or potentially remote control of the radar rig. The NUCBOX is also usefully small in physical size, with the dimensions listed in table 4.1 as 132mm x 125mm x 58mm. This eventually allows for the computing unit to be mounted to a rig, without taking up a disproportionately large amount of space. This is important because for testing purposes, having a smaller overall system makes testing said system dramatically easier. With all of this in mind, I am confident that the GMKtec NUCBOX is a good choice of equipment for this project.

4.3.1.2 Software Defined Radios

Once a computing unit has been chosen, the next most important, or some may argue, the most important component of the system, is the SDRs. The SDRs form the backbone of the entire transmit and receive RF tool-chain. Without them, the system cannot work at all. My exact choice of SDRs

¹At the time of writing this, Aug 2025

was somewhat limited by the following factors. Firstly, SDRs, especially research grade SDRs, can end up being fairly expensive.

I think I should add a little section here explaining the options that I had with regards to SDR choice (basically what SDRs the radar group has)

After assessing all the available options, I settled on the [National Instruments \(NI\)](#) Ettus N210 SDR. Specifically, I settled on the NI Ettus USRP N210 motherboard, paired with the NI SBX-40 daughter board. The NI Ettus USRP N210 has hot-swappable daughter board slots which allow for use with various daughter boards designed and manufactured by NI Ettus. Over the years, many daughter boards have been built by NI Ettus for various different purposes. The various examples of the different choices a consumer could face regarding the daughterboards available can be seen below in table 4.2. This information in the table is adapted from [17].

Daughterboard	Function	Low Frequency	High Frequency	Max RF Bandwidth per Channel
UBX ²	Transceiver	10 MHz	6 GHz	40 MHz 160 MHz
WBX ²	Transceiver	50 MHz	2.2 GHz	40 MHz 120 MHz
CBX ²	Transceiver	1.2 GHz	6.0 GHz	40 MHz 120 MHz
SBX ²	Transceiver	400 MHz	4.4 GHz	40 MHz 120 MHz
XCVR	Transceiver	2.4 GHz 4.9 GHz	2.5 GHz 5.9 GHz	36 MHz (RX) 48 MHz (TX)
Basic RX	Receiver	1 MHz	250 MHz	100 MHz
Basic TX	Transmitter	1 MHz	250 MHz	100 MHz
LFRX	Receiver	DC	30 MHz	30 MHz
LFTX	Transmitter	DC	30 MHz	30 MHz
DBSRX2	Receiver	800 MHz	2.3 GHz	60 MHz
TVRX2	Receiver	50 MHz	860 MHz	10 MHz

Table 4.2: Description of various commercially available daughterboards for the NI Ettus USRP N210

Owing to the fact that I want to build a radar system with two SDRs, one for handling transmit capabilities, and the other for handling receive capabilities, I needed two motherboards, and two daughterboards. Luckily for me, the UCT [Radar and Remote Sensing Group \(RRSG\)](#) has access to multiple USRP devices. Therefore, I settled on using the SBX-40 daughterboard. This daughterboard is a powerful multipurpose board with a very wide range of tunable frequencies, which is great because it allows for operating the radar system at multiple different center frequencies if needed. Pictured below is the SBX-40.

²The N-series family of USRPs is compatible only with 40 MHz bandwidth daughterboards, while the X-series can support both 40 MHz and 120/160 MHz daughterboards.

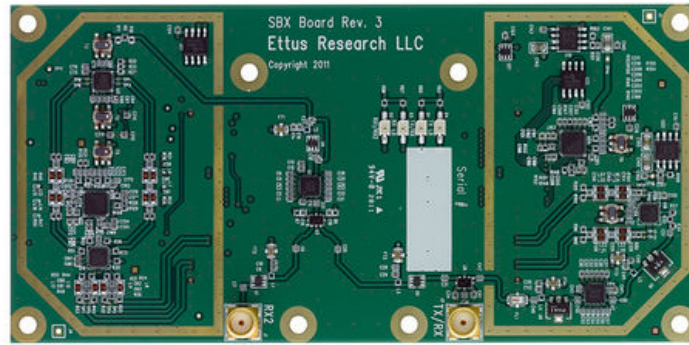


Figure 4.3: SBX-40 Daughterboard

Apart from the highlighted features in table 4.2 above, the SBX-40 has the following specifications listed below in table 4.3.

Frontends	2 quadrature frontends (1 transmit, 1 receive)
Local Oscillator	Independent receive and transmit LO's and synthesizers. Allows for full-duplex operation on different transmit and receive frequencies.
Antennas	Transmit: TX/RX Receive: TX/RX or RX2
Gain	Transmit Gains: PGA0, Range: 0-31.5dB Receive Gains: PGA0, Range: 0-31.5dB
Bandwidth	40 MHz, RX & TX
Sensors	lo_locked: boolean for LO lock state
MIMO Capability	Yes
LEDs	All LEDs flash when daughterboard control is initialized TX LD: Transmit Synthesizer Lock Detect TX/RX: Receiver on TX/RX antenna port (No TX) RX LD: Receive Synthesizer Lock Detect RX1/RX2: Receiver on RX2 antenna port
Freq Range	400MHz - 4.4GHz
Noise Figure	4 - 5 dB @ (400MHz ~1.5GHz) 5 - 7 dB @ (1.5GHz ~3GHz) 7 - 12dB @ (3GHz ~4.4GHz)
RX IIP3 (Max)	16 ~22 dBm
RX IQ Imbalance	-20 dBc @ (400MHz ~600MHz) -30dBc @ (600MHz ~4.4GHz)
TX Power (Max)	20 dBm @ (400MHz ~3.5GHz) 18 - 20dBm @ (3.5GHz ~4.4GHz)
TX OIP3	26 - 30dBm @ (400MHz ~1.5GHz) 30 dBm @ (1.5GHz ~4.4GHz)
TX IQ Imbalance	-20 dBc @ (400MHz ~600MHz) -30dBc @ (600MHz ~4.4GHz)
Input/Output Impedance	All RF Ports are matched to 50 Ohm with -10dB or better return loss generally.
Input Power Levels	The maximum input power for the SBX is -15 dBm.

Table 4.3: Specifications of SBX-40 Daughterboard

Since the SBX-40 meets all the required specifications that might be needed, particularly the fact that it is commercially available, it makes for a good choice to fall part of the SDR section of the system.

4.3.1.3 Antennas

The next component of the system which has to be chosen is the antennas. The antennas are important parts of the system as they directly handle any transmitted and received RF signals for the system as

a whole. Antennas play a crucial role in the performance of the system, and there are a lot of potential choices to be made. Much like the SDR choice that had to be made in the previous section, I made my decision of antenna based on the availability of antennas under control of the UCT RRSg. In lOu of this, I chose the H-1498 manufactured by [American Electric Laboratories \(AEL\)](#), partially because the total operating frequency range of the antenna falls within the desired range I have for my system, and partially because there were not too many other options available. AEL has changed ownership throughout the years, and is now owned by [Cobham Defense Systems \(CDS\)](#). This antenna has the following specifications.

Frequency Range	2-18 GHz
Gain	2 GHz = 6 dBiL
	9 GHz = 11 dBiL
	18 GHz = 11 dBiL
Power (Continuous Wave)	50 W

Table 4.4: H-1498 Antenna Specifications

As you can see, this antenna is a good choice for my project because the gain at the operating frequency of the system, roughly 2.7 GHz, is decent, and, more importantly, the antenna has been designed to operate from 2-18 GHz, which includes the frequency of the system. I am not an antenna design expert, and therefore I will not focus too much energy on the optimal antenna setup, and will rather focus my energy on the design of the system as a whole.

4.3.1.4 Carrying Rig

4.3.1.5 Periphery Devices

4.3.2 Software

4.4 Range Radar Pipeline Design

4.5 Communication Radar Pipeline Design

4.6 Range Radar Pipeline Design

Chapter 5

Validation and Simulation

5.1 Validation

5.1.1 Transmitter Validation

5.1.1.1 Power Module

5.1.1.2 Microcontroller

5.1.1.3 Sensor Modules

5.1.1.4 LoRa module and Display and Control Module

5.1.2 Receiver Validation

5.2 Simulation

Chapter 6

Methodology

In this chapter that follows, I will highlight the testing procedure that was followed for the system. Importantly, I will also explain the reasoning behind why I made the specific choices that I do as it pertains to the testing environment setup. The aim of the experiments detailed is to validate the previous system design work done for this thesis.

6.1 Acceptance Test Protocols

In light of the aforementioned requirements and specifications in section 4.1, the following [Acceptance Test Protocols \(ATPs\)](#) are derived to ensure that the specifications for the system have been met. The ATPs can be broken into three sections for the overall system, the range radar module, the communication module, and the classification module. The various ATPs for the different modules and the overall system are highlighted below. These ATPs have been derived in part using the specifications mentioned in section 4.1.2.

1. Overall System:

- (a) Ensure that the system follows the purchasing rules in regard to off the shelf scientific and consumer hardware outlined in specification 1(a).
- (b) Following from that, specification 1(b) should be achieved by testing whether the system can switch casually between modes of operation as described.

2. Range Radar Module:

- (a) The range resolution shall be measured by placing multiple different distance objects in a single frame to determine how accurately the radar can separate the objects based on their distances.
- (b) Object testing at distances greater than 10m must be performed.

3. Communication Module:

- (a) The communication module's performance shall be classified by testing the performance of the system over a wide range of SNR values. The system shall be characterized in a single graph outlining the performance at different SNR values.

- (b) The transmission scheme shall be analyzed to determine theoretical bit rates of the transmission system, ensuring that the transmission rate of the system satisfies specification 3(b), by making sure the transmission rate is at or above 1 Mbps.

4. Classification Module:

- (a) As outlined in specification 4(a), it must be shown that this classification module can successfully differentiate between a minimum of three targets in a scene.
- (b) Test the ability of the system to handle the classification of objects in motion.
- (c) Viability of object detection and classification at various distances away from the radar shall be examined.

6.2 Experimental Design

6.2.1 Overview

As mentioned in chapter 4, the overall radar system can be separated into three distinct modules. The overall system contains the range radar module, followed by the communication module, followed finally by the classification module. As mentioned previously, the first module involves the ability of the system to act as a simple range radar with the ability to determine the distance of object within a scene. The second module is involved with the ability of the overall system to communicate data from a transmission antenna to a reception antenna. The third and last module handles the classification ability of the radar with respect to the ability to differentiate between various targets in a scene. Due to this separation of modules, the testing of the system has been divided into three separate testing sections. Listed below are the various overall testing categories.

1. **Range Testing:** The purpose of the range test will be to determine if the RADAR system can operate as just that, a simple RADAR. The main aim is to determine if the system can detect objects at various distances from the transmitter and receiver setup in accordance with the specification and ATPs.
2. **Communication Testing:** This test is all about how well the system can operate as a bare-bones communication system. The reason I have decided to include this as a test is because, as mentioned before, my entire classification pipeline is built upon OFDM communication principles (namely the use of pilot symbols in determining channel estimation data), and since these OFDM principles are so vital to my project, it is fitting to make sure that the system can actually operate as a data transmission system. This is necessary to show, since the broad category of RADAR that this system falls into could be considered ‘communication RADAR’, or in essence, a RADAR that can use the same signal within communication operations and normal RADAR operations.
3. **Classification Testing:** This is the main testing that will take place for my RADAR system. This testing setup will thoroughly analyze the ability of the RADAR system to classify various targets in a scene. The tests will also try to determine the viability of this concept in a real world use case scenario.

Of these three major testing categories, where reasonably needed, it was attempted to perform each test indoors and outdoors. Following on from the above listed broad testing structures, the exact testing procedure of each testing category will now be explained below.

6.2.2 Communication Test

The testing procedure followed for this module has a main purpose of characterizing the performance of the communication channel for various SNR values. Doing this characterization will be paramount in determining if the communication system works properly and meets the specifications and ATPs defined for said module. Determining that the communication channel works properly has the by product effect of ensuring that the channel estimations used for the classification module are working as intended.

As mentioned in chapter 4, one frame of my OFDM based signal is 32768 samples long. That means that there are 4384 samples either side of my ‘signal block’ that have values equal to zero. These are my guard bands. That leaves 24000 samples for data and pilot symbols. I made my signal have 4000 pilots and 20000 data carriers. Based on this setup, I can transmit 40000 bits at a time. On the reception side, I am able to read ingest raw data, which I can then decode and get my transmitted bits out of. Bearing this in mind, since I have the bits that I transmitted, and the bits that I have received, I am able to compute a BER for the transmission. The performance of the system changes drastically as the overall [Signal to Noise Ratio \(SNR\)](#) of the transmission changes.

To try and manipulate the SNR of the transmission, I want to perform multiple tests at various amplification levels. At severely low amplification levels, one can deduce that there will be very little signal being transmitted through the system.

6.2.2.1 Test 1: Communication System Characterization via Loopback

Since my communication tool chain has the ability to calculate the SNR of each and every capture that it processes, test 1 will work with this understanding. I wanted to design a test that shows, as good as possible, the overall communication system performance. One way that this can be shown, is by measuring the performance of the system under different stress conditions. So, for a communication system, that would involve measuring the performance of the system under varying SNR conditions. A varying SNR is a good choice of system stress conditions to determine real world raw communication performance. This is because it simulates the real world conditions that most signals (at least those operating in a free spectral band) have to deal with.

Unfortunately, there is no way to directly control the SNR in a real world test like this, as opposed to a simulation. Therefore, it is necessary to control another aspect of the signal which directly affects the SNR. In this test’s case, the best way to do that is to control the level above the noise floor that the signal exists on the received side of the system. The best way to achieve this, is to attach an attenuator between the transmit and receive ports of the system. Then, from there, one can adjust the level of attenuation, which in turn, raises and lowers the level that the transmitted signal sits above the noise floor.

In my experience with the USRP-N210 paired with the SBX daughterboard, the noise floor tends to

hover around -100 dBV. The minimum level of attenuation that I have been using to test my system is roughly 15dB. Therefore, for this test of the communication, I will transmit the same communication signal through an attenuator, and step the attenuator from 100dB to 15dB of attenuation.

I have access to the following attenuators in my capacity as a UCT student, the Keysight 8496B, and the Keysight 8494B. These are pictured in figure 6.1 below:



(a) Keysight 8496B



(b) Keysight 8494B

Figure 6.1: Keysight Attenuators

While the attenuators pictured above come in the same package, they are anything but the same. The differences are listed below.

	Keysight 8496B	Keysight 8494B
Frequency Range	DC to 18 GHz	DC to 18 GHz
Steps	10 dB	1 dB
Attenuation Range	0 dB - 110 dB	0 dB - 11 dB

Table 6.1: Keysight 8496B and 8494B Attenuator Characteristics

Importantly, both attenuators operate from 0 Hz to 18 GHz, which is more than enough bandwidth for testing this system, since this system is limited to operations within the bandwidth of the SBX daughterboard of the USRP-N210, which can transmit signals on a carrier wave of 400 MHz to 4.4 GHz. More over, during the system design, I settled on doing tests in the portion of the spectrum from around 2 GHz to 3 GHz.

Using both attenuators, daisy-chained together, gave me the ability to control attenuation in the range from 0 dB to 121 dB in steps of 1 dB. The setup for this loopback test can be seen in the diagram presented in fig 6.2 below:

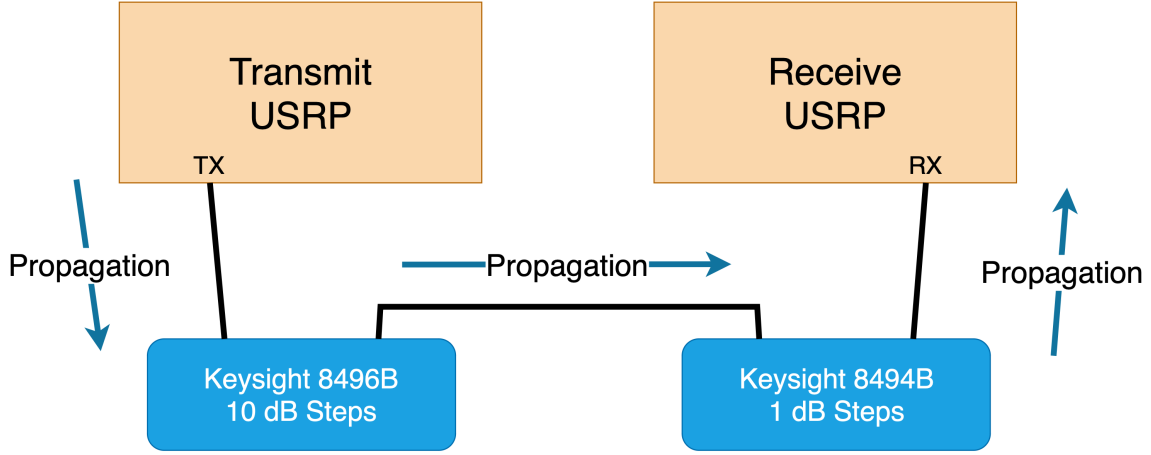


Figure 6.2: Communication Loopback Test

With this testing setup, as mentioned before, I have control over the attenuation of the overall signal propagation, effectively giving me indirect control over how high above the noise floor the transmitted signal sits, or in other word, the SNR. For the test, I will transmit a known communication signal, with known bits embedded into the signal. On the reception side, I can get a channel estimation, equalize the signal and demodulate the signal to extract data bits from the signal. I will do this for each attenuation level, varying from 100 dB to 15 dB of attenuation. For each signal that is received and demodulated, I am able to get an SNR and a BER. To ensure thorough characterization of the communication tool chain, I will perform 50 transmissions for each attenuation level. I will take all 50 SNR and BER values per attenuation level, and average them. This should give some very good insight into the performance of the system. The raw data format of the extracted data from this test will be in the format as shown below in table 6.2:

Attenuation	SNR_1 (dB)	BER_1 (%)	.	SNR_{50} (dB)	BER_{50} (%)	$Avg.SNR$ (dB)	$Avg.BER$ (%)
100 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
99 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
.	x_1	y_1	.	x_{50}	y_{50}	$\bar{x} = \frac{1}{n} \sum_{i=1}^{50} x_i$	$\bar{y} = \frac{1}{n} \sum_{i=1}^{50} y_i$
16 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
15 dB	x_1	y_1	.	x_{50}	y_{50}	.	.

Table 6.2: Communication Test 1 Data Format

Something that must be noted however, is that since my system calculates an SNR value for a received signal after said signal has propagated through a noisy environment, it will struggle to measure the true SNR of the signal below a certain attenuation level. My system will not be able to truly calculate the SNR from the received signal when the embedded signal is below the noise floor. This is not such a large problem however, since one can determine the SNR values achieved for low values of SNR by extrapolating the values at high SNR values to the low SNR values. This can only be done because we are in complete control of the attenuation level of the loopback system.

6.2.2.2 Test 2: Long Cable Loopback Test

After completing a complete communication tool-chain characterization test as outlined in the previous section, it is now important to show that the system can actually perform it's required tasks in the real world. This entails two more tests. However, it must be noted that the next two tests will not be as thorough as test 1, since test 1 mainly dealt with characterization of the communication system. Test 2 and 3 will primarily to ascertain whether or not the system can operate under non-ideal (real-world) conditions. Test 2 will be another loopback test, however this one will make use of a particularly long coaxial cable. This allows one to examine the performance of the communication system in a slightly more real world operational conditions, namely, the fact that the coaxial is exposed to more real-world environmental noise since it is longer than the shorter loopback test.

6.2.2.3 Test 3: Real World Antenna Test

The third and final test will involve a demonstration that the system can work while transmitting data over a real-world channel, namely, that of the air between two separate antennas. The goal with this test, much like the second test, will be to demonstrate that the system can indeed work in non-ideal conditions.

6.2.3 Range Test Tests

The next tests which must be performed are the tests that will help characterize the system's ability to capture range information about targets in a scene of interest. Due to previous design choices, the RADAR system does not have the ability to determine velocity of targets in a scene, and so for that reason, the range tests will be reserved to simple distance calculations of objects in a scene.

6.2.3.1 Test 1:

6.2.4 Classification Test

[1]

Chapter 7

Results

In the following section, the results obtained by performing the experiments described in chapter 6 will be presented. Minimal discussion for the presented results will be discussed, as discussion will be predominantly saved for chapter 8. The following results were obtained for each of the tests performed.

7.1 Communication Testing

7.1.1 Test 1: Communication System Characterization via Loopback

The following graph can be compiled for the results obtained.

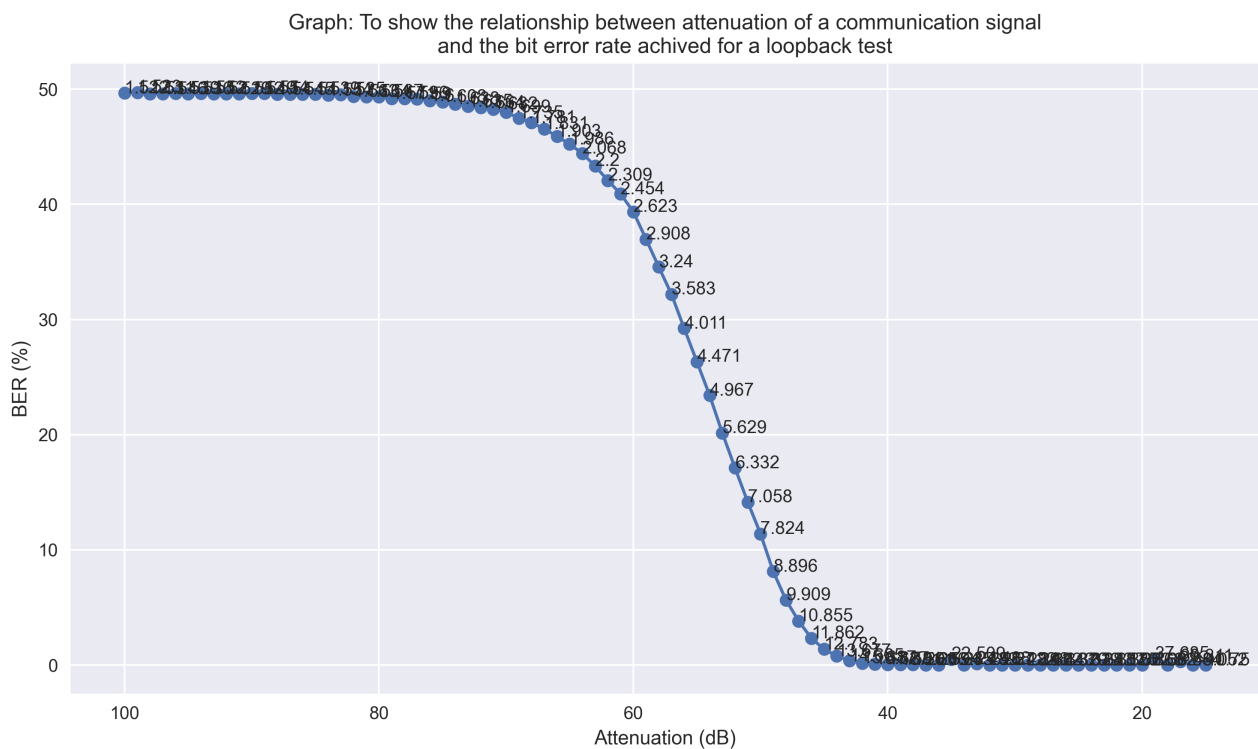


Figure 7.1: Keysight 8496B

7.2 100m Test

7.3 300m Test

7.4 500m Test

7.5 20km Test

7.6 Average RSSI Values with Distance

Chapter 8

Discussion

[8]

8.1 RSSI Values with Distance

8.2 Overall Testing Performance

8.3 Determining the Optimal Packet Structure

8.4 Determining the Optimal LoRa Parameters

Chapter 9

Conclusion and Recommendations

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Chapter 10

Appendix

10.1 Full Transmitter Circuit Schematic

10.2 Full Receiver Circuit Schematic

10.3 Long Range Test Interference

10.4 GitHub Link

10.5 Ethics Clearance